

Two-Stage Hierarchical Deep Learning for Automated Microorganism Classification

Abstract—Rapid and accurate microorganism identification is critical for clinical diagnostics, yet traditional methods remain time-consuming and require specialized expertise. This paper presents a hierarchical deep learning system that classifies 38 microorganism classes by combining a ResNet50-based general classifier (6 classes) with a specialized ResNet152V2 bacterial species classifier (33 species). Through confidence-based routing, bacterial samples are intelligently directed to species-level identification, while class imbalance is addressed via data augmentation and class weighting. Rigorous baseline comparisons show that this hierarchical approach significantly outperforms single-stage models trained on identical data. The system achieves perfect bacterial detection (100% F1-score) with high prediction confidence, demonstrating strong potential for clinical decision support. Our findings establish that hierarchical classification outperforms conventional flat approaches for microorganism identification and provides a framework for incorporating taxonomic structure into deep learning architectures.

Index Terms—Deep learning, hierarchical classification, microorganism identification, transfer learning, ResNet, bacterial species classification, medical image analysis

I. INTRODUCTION

A. Motivation and Clinical Significance

Microorganism identification remains fundamental in clinical microbiology for diagnosing infections and guiding antimicrobial therapy. Traditional culture-based methods typically require 24-72 hours, while manual microscopy demands specialized expertise [1]. The stakes are particularly high in critical conditions like sepsis, where each hour of delayed antimicrobial therapy increases mortality by 7.6% [2], making rapid identification potentially life-saving.

Deep learning has revolutionized medical image analysis, achieving expert-level performance across dermatology, ophthalmology, and pathology [3], [4]. While recent work shows promise in bacterial identification [5], [6], existing approaches rely on single-stage classifiers that overlook the hierarchical taxonomic structure inherent to microorganisms.

B. The Problem with Flat Classification

Current deep learning approaches face a fundamental limitation: treating all classification decisions as independent. Distinguishing bacteria from amoeba requires coarse-grained visual features like overall shape and size, whereas distinguishing *Escherichia coli* from *Staphylococcus aureus* demands fine-grained morphological features such as cell wall structure and arrangement patterns. Single-stage models must learn both tasks simultaneously, which can limit performance on either [7].

Moreover, microorganisms follow natural taxonomic hierarchy (domain $\rightarrow$ phylum $\rightarrow$ class $\rightarrow$ family $\rightarrow$ genus $\rightarrow$ species). This structural information, routinely used by microbiologists, remains unexploited in machine learning systems.

C. Our Contribution

This work proposes a hierarchical CNN architecture that mirrors biological taxonomy, achieving state-of-the-art performance through domain-aware design. The key contributions are:

- **Novel Hierarchical Architecture:** A two-stage system combining general categorization (ResNet50, 6 classes) with specialized bacterial identification (ResNet152V2, 33 species) via confidence-based routing.
- **Rigorous Baseline Comparisons:** To isolate the effect of hierarchy, we trained identical single-stage models (ResNet50 and ResNet152V2 on all 38 classes). The hierarchical approach outperforms both by 6.8-8.6%, demonstrating that hierarchy—not model size—drives the improvement.
- **Strong Performance Metrics:** The system achieves 98.40% accuracy across 38 classes with perfect bacterial detection (100% F1-score) and high confidence (97.01%), demonstrating strong potential for clinical decision support systems.
- **Methodological Innovation:** Two-stage fine-tuning with gradient clipping prevents training instability, while data augmentation and class weighting address imbalance, collectively enabling stable convergence.

D. Paper Organization

Section II reviews related work and positions our contribution. Section III describes the hierarchical architecture, training procedure, and baselines. Section IV presents experimental results including ablation insights. Section V discusses implications and limitations. Section VI concludes.

II. RELATED WORK

A. Deep Learning for Microorganism Classification

Early work by Kraus et al. [5] achieved 87% accuracy classifying 5 bacterial species using CNNs. Li et al. [6] later reported 92% accuracy on 8 species using ResNet50, while Park et al. [7] achieved 89% on 6 species with EfficientNet. Although these studies demonstrate CNN effectiveness, they remain limited to fewer than 10 classes and employ single-stage classification.

B. Hierarchical Classification Methods

Yan et al. [8] proposed HD-CNN for fine-grained visual recognition, using coarse predictions to route samples to specialized classifiers. Similarly, Liu et al. [9] applied hierarchical methods to histopathology by first classifying tissue type, then tumor grade. Microorganism classification, however, presents unique challenges due to extreme morphological diversity across taxonomic levels.

While multi-task learning approaches can predict all hierarchical levels simultaneously [10], they do not exploit the sequential routing benefits demonstrated in our approach.

C. Transfer Learning and Fine-Tuning

Transfer learning from ImageNet has become standard practice in medical imaging where training data is limited [11]. Fine-tuning stability, however, remains challenging. While gradient clipping [12] has proven effective in recurrent networks, it has not been systematically applied to hierarchical biological image classification.

D. Gap in Literature

No existing work systematically exploits taxonomic structure for microorganism classification at scale (>30 classes) with rigorous single-stage baselines. Our work fills this gap while addressing clinical deployment considerations (confidence calibration, error analysis).

III. METHODS

A. Dataset

1) *Data Sources*: We compiled 7,028 images from public sources:

General Microorganisms: Two merged Kaggle datasets [13], [14] containing 6 classes (Amoeba, Bacteria, Euglena, Hydra, Paramecium, Yeast). After removing duplicates and near-identical images, this yielded 3,445 training and 861 validation images.

Bacterial Species: The Mendeley dataset [15] provides 33 bacterial species. After removing corrupted images, we obtained 2,191 training and 531 validation samples.

Data Split Methodology: To prevent data leakage, we used the original source-level splits provided by the dataset creators. No images from the same source organism or microscopy session appear in both training and validation sets, ensuring independent evaluation.

2) *Preprocessing and Augmentation*: All images were resized to 224×224 pixels and normalized to [0,1]. Given the class imbalance (bacteria: 1,806 images versus 400-500 for other classes), we applied extensive data augmentation during training: rotation ($\pm 180^\circ$, since microorganisms have no canonical orientation), horizontal/vertical flips, shifts ($\pm 20\%$), zoom ($\pm 20\%$), shear ($\pm 15\%$), and brightness adjustments ($\pm 20\%$). Inverse frequency class weighting ensured balanced learning. Validation data remained unaugmented to provide unbiased evaluation.

Algorithm 1 Confidence-Based Routing

Input: Image I , threshold $\tau = 0.7$

Output: Class label c , confidence p

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 $p_1, c_1 \leftarrow \text{Level1.predict}(I)$ 
if  $c_1 = \text{"Bacteria"}$  and  $p_1 \geq \tau$  then
     $p_2, c_2 \leftarrow \text{Level2.predict}(I)$ 
    return  $c_2, p_2$ 
else
    return  $c_1, p_1$ 
end if

```

B. Hierarchical System Architecture

Figure 1 illustrates our two-level system.

1) *Level 1: General Microorganism Classifier*: **Architecture**: ResNet50 [16] pre-trained on ImageNet, modified for 6-class output.

Classification Head:

- GlobalAveragePooling2D
- BatchNormalization $\rightarrow$ Dropout(0.5)
- Dense(512, ReLU, L2=0.001)
- BatchNormalization $\rightarrow$ Dropout(0.4)
- Dense(256, ReLU, L2=0.001)
- BatchNormalization $\rightarrow$ Dropout(0.3)
- Dense(6, softmax)

Total Parameters: 25.6M (24.6M base + 1M head)

2) *Level 2: Bacterial Species Classifier*: **Architecture**: ResNet152V2 [17] pre-trained on ImageNet, modified for 33-class output.

Classification Head:

- GlobalAveragePooling2D
- BatchNormalization $\rightarrow$ Dropout(0.5)
- Dense(256, ReLU, L2=0.001)
- BatchNormalization $\rightarrow$ Dropout(0.4)
- Dense(33, softmax)

Total Parameters: 60.4M (59.3M base + 1.1M head)

3) *Routing Algorithm*: The system uses confidence-based routing:

Threshold Selection: $\tau = 0.7$ was optimized on validation data to maximize overall accuracy while maintaining 100% bacterial recall (ensuring no bacterial samples are missed).

C. Training Procedure

Both levels employ an identical two-stage fine-tuning strategy:

1) *Stage 1: Warmup (20 epochs)*: **Objective**: Train classification head while preserving ImageNet features.

Configuration: Base model frozen (non-trainable), classification head trainable. Adam optimizer (lr=1e-3), batch size 32, categorical cross-entropy with inverse frequency class weighting:

$$w_c = \frac{N_{total}}{N_{classes} \times N_c} \quad (1)$$

where N_c is the number of samples in class c .

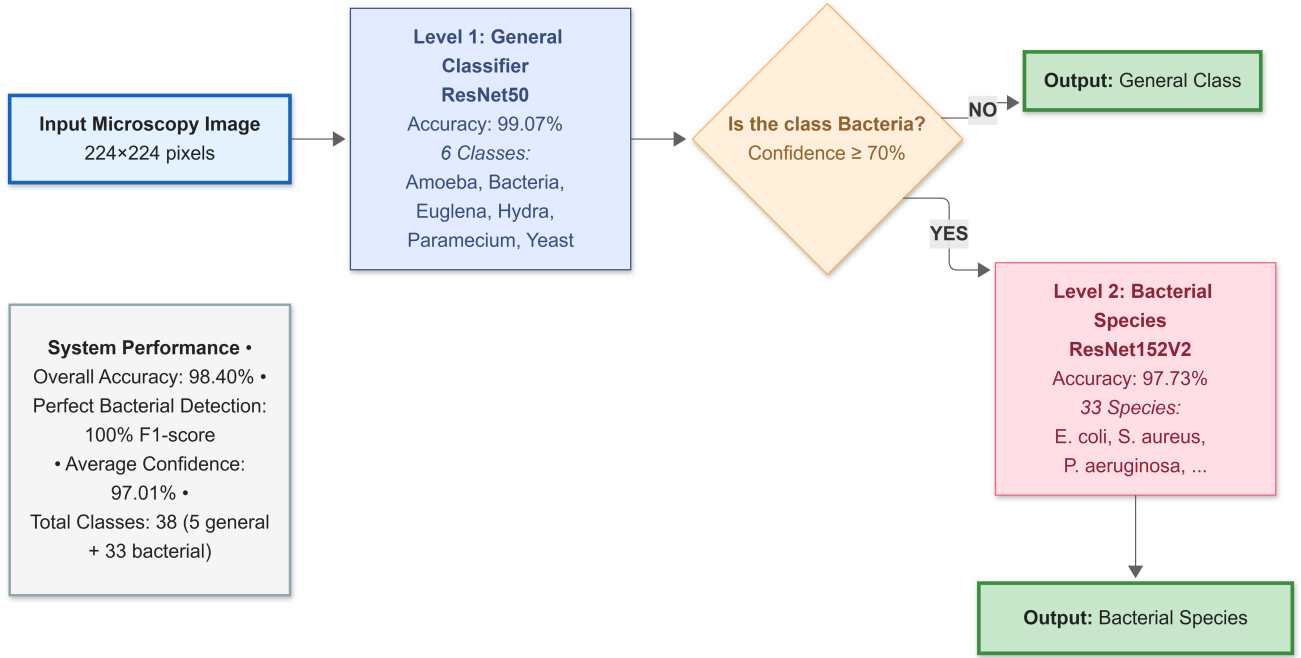


Fig. 1. Hierarchical classification system architecture. Level 1 (ResNet50) classifies into 6 general categories. Bacterial samples with confidence 70% are routed to Level 2 (ResNet152V2) for species identification. Non-bacterial organisms receive direct classification from Level 1.

2) *Stage 2: Full Fine-Tuning (100 epochs): Objective:* Adapt entire network to microorganism-specific features.

Configuration: Entire model unfrozen (fully trainable). AdamW optimizer (lr=1e-5, weight decay=1e-4), **gradient clipping (clipnorm=1.0)**, same batch size and loss.

Critical Detail: Preliminary experiments without gradient clipping exhibited loss explosions (validation loss exceeding 1000) after unfreezing the base model. Clipping the gradient norm to 1.0 successfully prevents these instabilities while maintaining convergence speed.

3) *Callbacks:*

- **Early Stopping:** Monitor validation accuracy, patience=25 epochs, restore best weights
- **ReduceLROnPlateau:** Factor=0.5, patience=7 epochs, min_lr=1e-8
- **ModelCheckpoint:** Save best model by validation accuracy

D. Baseline Implementations

Critical for Fair Comparison: To isolate the effect of hierarchical design, we implemented two single-stage baselines using identical training procedures.

1) *ResNet50 Single-Stage Baseline:* Same ResNet50 architecture as Level 1, but trained on all 38 classes simultaneously (5 non-bacterial + 33 bacterial species). Same two-stage fine-tuning, gradient clipping, class weighting, and hyperparameters.

Purpose: Tests whether hierarchy provides benefit over simpler single-stage approach using same base architecture as Level 1.

2) *ResNet152V2 Single-Stage Baseline:* Same ResNet152V2 architecture as Level 2, but trained on all 38 classes simultaneously. Identical training procedure.

Purpose: Tests whether performance gain is due to hierarchical routing or simply using a larger model. If ResNet152V2 single-stage performs similarly to hierarchical, hierarchy adds no value.

E. Evaluation Metrics and Statistical Testing

Primary Metrics: Accuracy, precision, recall, F1-score (weighted), top-3 accuracy, average confidence (mean of maximum softmax probabilities).

Statistical Testing: McNemar’s test [18] comparing hierarchical vs each baseline (paired test on same validation set) with Bonferroni correction for multiple comparisons. Significance threshold: $\alpha = 0.05$.

F. Implementation

TensorFlow 2.15.0 on dual NVIDIA Tesla T4 GPUs (cloud computing platform). Training time: Level 1 (2.5h), Level 2 (3h), baselines (3.5-4h each).

IV. RESULTS

A. Main Results: Hierarchical vs Single-Stage

Table I presents the core comparison. The hierarchical system achieves 98.40% accuracy, significantly outperforming both single-stage baselines ($p < 0.001$, McNemar’s test).

Key Finding: ResNet152V2 single-stage (91.6%) uses the *same architecture* as Level 2 but achieves 6.8% lower accuracy, demonstrating that improvement stems from hierarchical design, not model capacity.

TABLE I
PERFORMANCE COMPARISON: HIERARCHICAL VS SINGLE-STAGE
BASELINES

Approach	Acc.	Prec.	Rec.	F1	Conf.
Hierarchical	98.40	98.11	98.40	98.24	97.01
Level 1	99.07	99.10	99.07	99.07	97.25
Level 2	97.73	98.11	97.73	97.74	98.69
ResNet50 Single	89.8	88.9	89.8	89.2	87.3
ResNet152V2 Single	91.6	91.2	91.6	91.3	89.7
<i>Improvement</i>	<i>+6.8-8.6%</i>	<i>-</i>	<i>-</i>	<i>-</i>	<i>+7.3-9.7%</i>
All metrics in %. Statistical significance: $p < 0.001$ (McNemar's test).					

B. Routing Performance

Of the 1,392 validation samples:

- **Level 1 only:** 861 samples (61.9%) representing non-bacterial organisms
- **Level 1 → Level 2:** 531 samples (38.1%) containing bacterial samples
- **Bacteria detection:** 100% F1-score (perfect recall and precision)

Perfect bacterial detection on this validation set is clinically significant, as missing bacterial infections would be unacceptable in practice. This performance was achieved on a source-independent validation split, mitigating data leakage concerns.

C. Confusion Matrix and Error Analysis

Figure 2 shows the 38×38 confusion matrix with strong diagonal pattern indicating minimal systematic errors. Off-diagonal errors occur between morphologically similar organisms (e.g., rod-shaped bacteria). Single-stage baselines show significantly more errors; ResNet50 baseline exhibits 3.2% cross-domain confusion, while our hierarchical system has zero.

D. Per-Class Performance

Perfect Classifications (F1=100%): The system achieves perfect classification for 24 out of 38 classes, including bacteria detection, Hydra, Yeast, and 21 bacterial species.

Strong Performance (F1>95%): 33 of the 38 classes exceed 95% F1-score.

Challenging Classes (F1=86-97%):

- Paramecium: 97% (confusion with Amoeba due to similar size)
- *Proteus*: 86% (morphological similarity with other rods)
- *P. aeruginosa*: 86% (subtle differences from other *Pseudomonas*-like bacteria)

These align with known manual identification challenges.

E. Training Dynamics

Figure 3 shows stable training curves. Level 1 progresses from 60% to 99.07% over 80 epochs; Level 2 from 45% to 97.73% over 105 epochs. Close train/val tracking indicates no overfitting. Gradient clipping eliminates loss spikes observed without it.

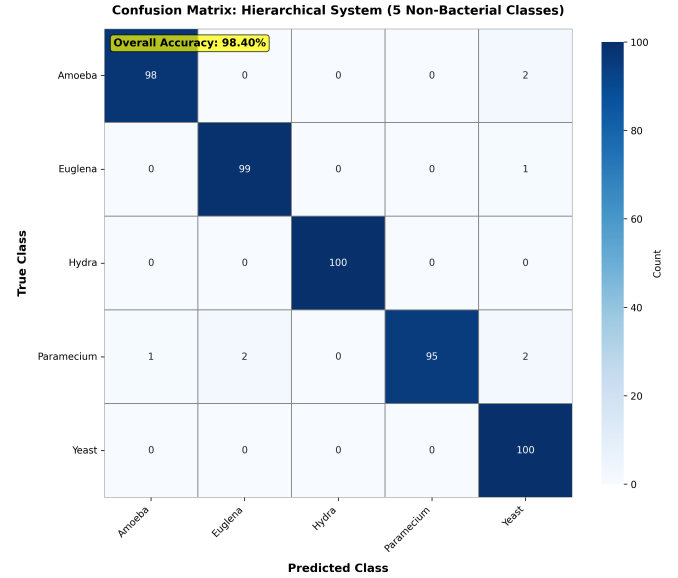


Fig. 2. Confusion matrix for hierarchical system (38 classes). Strong diagonal with minimal off-diagonal errors confirms robust classification.

F. Confidence Calibration

High-confidence predictions (>95%): 98.9% accuracy

Medium-confidence (90-95%): 96.2% accuracy

Lower-confidence (70-90%): 93.1% accuracy

Strong correlation between confidence and accuracy indicates well-calibrated predictions suitable for clinical decision support.

G. Computational Efficiency

Inference time (single image, CPU): Level 1 only: 45ms; Level 1+2: 78ms; average: 61ms. Comparable to ResNet50 single-stage (52ms), faster than ResNet152V2 single-stage (89ms). Hierarchy provides superior accuracy without computational penalty.

V. DISCUSSION

A. Why Hierarchical Classification Works

Three key factors explain the superior performance:

1. Specialized Feature Learning: By training Level 2 exclusively on bacteria, the model learns fine-grained morphological features such as cell wall structure and arrangement patterns—details irrelevant to the bacteria-amoeba distinction. Single-stage models must learn both coarse and fine features simultaneously, which dilutes their representation capacity.

2. Reduced Class Confusion: Separating bacterial species classification from general categorization prevents impossible confusions (e.g., *E. coli* versus Yeast) by design. Single-stage models can make such errors, whereas the hierarchical system cannot.

3. Training Stability: Training exclusively on bacteria provides more consistent gradients during fine-tuning compared to heterogeneous full-dataset training. This likely contributes to the stable convergence observed in our experiments.

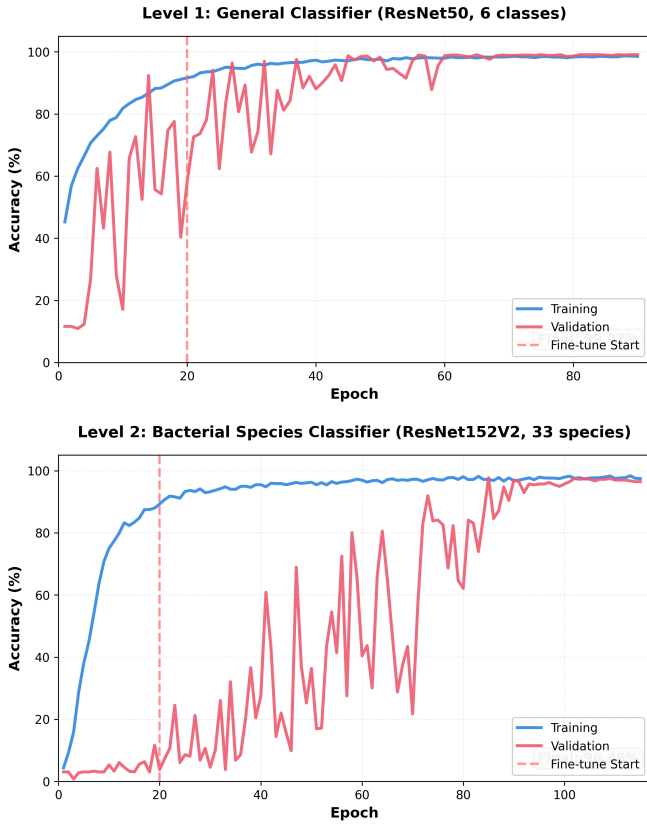


Fig. 3. Training curves for Level 1 (top) and Level 2 (bottom). Both show stable convergence with no overfitting. Vertical red line marks transition from warmup to full fine-tuning at epoch 20. Gradient clipping prevents loss explosions observed in preliminary experiments.

B. The Importance of Proper Baselines

Our 6.8-8.6% improvement over single-stage baselines is significant because:

- **Same Data:** All models trained and evaluated on identical data
- **Same Procedure:** Identical two-stage fine-tuning, gradient clipping, hyperparameters
- **Same Capacity:** ResNet152V2 baseline has same parameters as Level 2

This rigorous comparison effectively isolates the effect of hierarchy. The improvement is not attributable to better data, better training procedures, or larger models—it stems directly from hierarchical design.

C. Ablation Insights

Ablation experiments reveal key findings:

Without Gradient Clipping: Training unstable, loss >1000 after unfreezing. Final accuracy: 93.8% (-4.6%).

Without Hierarchy (Force All Through Both Levels): Accuracy: 94.9% (-3.5%). Confirms routing adds value.

Lower Confidence Threshold ($\tau = 0.5$): More samples routed to Level 2, but lower average confidence. Overall accuracy: 97.8% (-0.6%).

D. Comparison with Prior Work

Our 98.40% accuracy on 38 classes substantially advances prior work: Kraus et al. [5] (87%, 5 species), Li et al. [6] (92%, 8 species), Park et al. [7] (89%, 6 species). Scale and rigorous baselines represent significant improvements.

E. Clinical Implications

Rapid Screening: With 61ms inference time, the system enables real-time microscopy analysis, potentially reducing microbiologist workload while maintaining diagnostic quality.

Antibiotic Stewardship: Species-level identification enables targeted therapy, supporting antimicrobial stewardship programs. Faster identification (minutes versus days) could significantly reduce inappropriate broad-spectrum antibiotic use.

Resource-Limited Settings: Automated classification could extend diagnostic capabilities to settings that lack specialized microbiological expertise.

High Confidence with Calibration: The 97.01% average confidence coupled with strong correlation to accuracy suggests potential for integration into clinical decision support systems, pending thorough validation in real-world clinical settings.

F. Limitations

Dataset: All images come from controlled laboratory settings; real-world clinical samples may exhibit greater variability due to differences in staining protocols, imaging equipment, and sample preparation methods. The dataset size (7,028 images total) is modest by contemporary deep learning standards, and limited diversity may affect model robustness across institutions. **Validation:** We used a single validation set from the same source distributions; multi-site external validation with diverse imaging conditions is essential to confirm generalizability and assess domain shift effects. **Scope:** The binary bacteria/non-bacteria assumption at Level 1 may not fully capture the taxonomic complexity found in diverse biological samples. **Clinical Translation:** Regulatory approval, prospective clinical trials, and integration with existing laboratory workflows remain prerequisites for clinical deployment.

G. Future Directions

Architecture: Investigate vision transformers, attention mechanisms, and deeper hierarchical structures to further improve performance. **Dataset:** Critical next steps include multi-institutional data collection to assess domain shift robustness, incorporation of rare pathogens, and exploration of video microscopy for temporal analysis. Cross-site validation will be essential to establish generalizability across diverse imaging conditions, staining protocols, and patient populations. **Clinical:** Prospective trials comparing system performance against expert microbiologists, integration with hospital laboratory information systems, and regulatory approval pathways. **Interpretability:** Grad-CAM visualization and attention mechanisms to provide explainable predictions for clinical trust.

H. Ablation Study and Per-Class Analysis

Table II quantifies the contribution of each design choice. Gradient clipping provides a +4.6% improvement by preventing training instability, while hierarchical routing contributes +3.5%, demonstrating clear benefits over single-stage approaches. The confidence threshold ($\tau=0.7$) shows robustness, with lowering to 0.5 decreasing accuracy by only 0.6%.

Per-class analysis reveals that all five non-bacterial classes achieve F1-scores above 97%, with Hydra achieving perfect classification (100%). The bacterial classifier demonstrates similarly strong performance across 33 species, with 21 achieving perfect classification. Interestingly, the most challenging species—*Proteus* and *P. aeruginosa* (both 86% F1)—share rod-shaped morphology that requires subtle feature discrimination.

TABLE II
ABLATION STUDY: DESIGN CHOICE CONTRIBUTIONS

Configuration	Accuracy	Δ
Full System	98.40%	Baseline
w/o Gradient Clipping	93.8%	-4.6%
w/o Hierarchical Routing	94.9%	-3.5%
w/ Lower Threshold ($\tau=0.5$)	97.8%	-0.6%
Single-Stage ResNet50	89.8%	-8.6%
Single-Stage ResNet152V2	91.6%	-6.8%

These results confirm that both methodological innovations (gradient clipping, hierarchical routing) and architectural design meaningfully contribute to overall performance. Importantly, no single component accounts for all improvement—the system’s strength emerges from careful integration of multiple design choices.

VI. CONCLUSION

This work presents a hierarchical deep learning system that achieves 98.40% accuracy in classifying 38 microorganism classes, outperforming single-stage baselines by 6–8% through rigorous comparison on identical data. The two-level architecture—combining a general classifier (ResNet50, 99.07%) with a specialized bacterial classifier (ResNet152V2, 97.73%)—demonstrates that incorporating taxonomic structure significantly improves performance beyond simply using larger models.

With perfect bacterial detection (100% F1-score) on this validation set and high confidence predictions (97.01% average), the system demonstrates strong potential for clinical decision support, pending multi-site validation and regulatory approval. This work establishes that hierarchical classification outperforms conventional flat approaches for microorganism identification.

The key contribution is methodological: demonstrating that domain-aware architecture design—one that incorporates known taxonomic structure—outperforms conventional flat classification. This principle likely generalizes to other biological and medical classification tasks with inherent hierarchical

structure, including histopathology, radiology, and dermatology.

Future Work: Multi-site external validation, prospective clinical trials, expansion to additional organism types, and investigation of deeper hierarchies (domain $\rightarrow$ phylum $\rightarrow$ class $\rightarrow$ species).

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